Master's Thesis

Multi-Agent Reinforcement Learning for Joint Interference Avoidance and Gateway Load Balancing via Beam Hopping in NTN

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Abstract

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List of Abbreviations

3GPP	3rd Generation Partnership Project
ACK	Acknowledgment
BHO	Basic Handover
CHO	Conditional Handover
EMC	Earth-Moving Cell
FSPL	Free Space Path Loss
GPS	Global Positioning System
GW	Gateway
HO	Handover
HOPP	Ping-Pong Handover
ISD	Inter-Site Distance
KPI	Key Performance Indicator
LEO	Low Earth Orbit
MR	Measurement Report
NTN	Non-Terrestrial Networks
OID	Orbital Information and Distance
RACH	Random Access Channel

RB	Resource Block
RLF	Radio Link Failure
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SIB	System Information Block
SINR	Signal-to-Interference-plus-Noise Ratio
TN	Terrestrial Networks
ToS	Time of Stay
TTT	Time-to-Trigger
UE	User Equipment
UHO	Unnecessary Handover

Chapter 1. Introduction

1.1. Background and motivation

LEO satellite communication has recently emerged as a promising next-generation technology for providing global coverage. The architecture of an LEO network consists of a multitude of satellites concurrently serving numerous terrestrial cells. However, this dense multi-satellite and multi-cell environment can lead to significant inter-cell interference. Furthermore, satellites require stable and high-capacity backhaul connections with GWs to prevent data traffic bottlenecks and ensure seamless service for the cells they cover.

One of the most critical challenges in LEO satellite communications stems from the high mobility of the satellites. As satellites orbit the Earth at extremely high speeds, a ground terminal must frequently change its serving satellite. This high mobility necessitates frequent handovers and switchovers, which become a primary cause of degraded service continuity and can lead to temporary network disconnections.

In particular, conventional handover schemes based on received signal strength exhibit several limitations in this dynamic LEO environment.

Such schemes, which simply switch the connection to the satellite with the strongest signal, can trigger unnecessarily frequent handovers. Moreover, they often lead to the ping-pong effect, where handovers are rapidly repeated back and forth between two satellites at the cell edge. Consequently, the terminal's ToS with a single satellite becomes very short, which significantly degrades overall network efficiency and stability.

Therefore, to guarantee stable QoS and service continuity in LEO satellite networks, it is essential to develop intelligent handover and switchover schemes that can effectively manage the challenges posed by satellite mobility. This paper proposes a novel approach to address these pressing issues.

1.2. Contributions

To address the challenges of high mobility and dynamic network topology in LEO satellite networks, this paper proposes a novel handover and resource management scheme based on multi-agent reinforcement learning. The key contributions of this work are summarized as follows:

- **A MARL Framework for Joint Optimization:** We design a MARL framework where each satellite acts as an autonomous agent. These agents learn collaboratively to find the optimal policy for selecting both user cells and gateways simultaneously, adapting to the highly dynamic network environment without the need for a centralized controller.

- **User Link Stability by Minimizing Handovers while Maximizing SINR:** Our proposed scheme enables the satellite agents to select user cells with a sophisticated dual objective. It aims to **maximize the SINR** for all served cells while concurrently **minimizing the total number of handovers**. This approach directly mitigates the ping-pong effect and enhances service stability for ground users.

- **Feeder Link Reliability by Preventing Bottlenecks and Reducing Switchovers:** The agents intelligently select gateways to **prevent traffic bottlenecks** in the backhaul network, ensuring efficient data routing. Crucially, this selection is performed while also **minimizing the frequency of feeder link switchovers**, which enhances the stability and reliability of the satellite-to-ground connection.

The effectiveness of our proposed scheme is validated through extensive simulations, which demonstrate significant improvements in network stability and user QoS compared to conventional, signal-strength-based methods.

1.3. Overview

The remainder of this thesis is structured as follows: Sec. II discusses handover, feeder link switching, and related research in beam hopping and resource allocation. Sec. III presents the proposed multi-agent re-

inforcement learning framework in detail. Sec. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Sec. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite HO and FLSO schemes

In an NTN environment, the continuity of the service link between the UE and the satellite is ensured through HO. HO is the procedure in Fig. 2.1 by which a UE changes the serving cell. Generally, this network-controlled procedure consists of three main phases: preparation, execution, and completion. Initially, the source gNB configures the UE to perform measurements on neighboring cells, and the UE sends a MR back to the network. Based on this report and other RRM information, the source gNB decides to initiate a HO and sends a HO REQUEST to a target gNB. The target gNB performs admission control, prepares resources, and responds with a HO REQUEST ACKNOWLEDGE. The source gNB then commands the UE to switch to the target cell by sending an RRCReconfiguration message. Upon receiving this, the UE synchronizes with the target cell and sends an RRCReconfigurationComplete message to finalize the process. Subsequently, the target gNB requests the core network to switch the data path and triggers the release of the UE's resources at the source gNB. However, applying this procedure in NTNs

presents unique challenges. Unlike in terrestrial networks, the difference in received signal strength between satellites is often small, making traditional measurement-based triggers less reliable. Consequently, distance- or time-based HO criteria are also employed to manage the predictable mobility of satellites. In addition to the service link, the continuity of the feeder link connecting the satellite to the ground GW is equally important. The presence or absence of a gNB onboard the satellite determines whether a feeder link switchover impacts the service link. For a transparent satellite without an onboard gNB, a feeder link switchover changes the serving GW on the ground, which in turn changes the serving gNB, resulting in a necessary HO procedure for the UE. In contrast, for a regenerative satellite equipped with an onboard gNB, a feeder link switchover can be handled at the transport layer without changing the serving gNB, making the switch transparent to the UE and avoiding an HO.

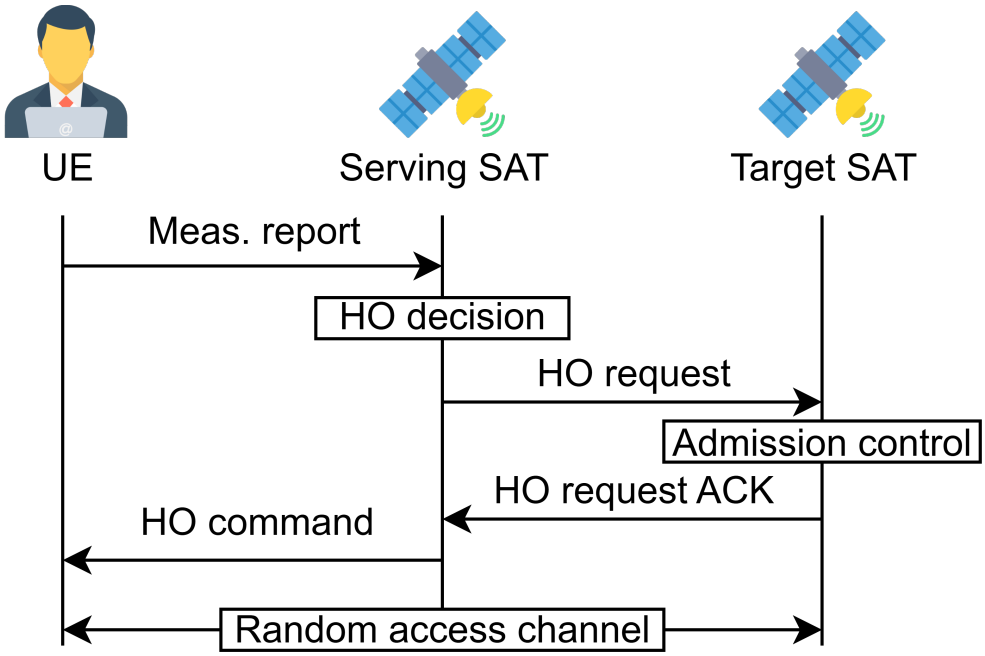


Fig. 2.1. Basic HO procedure

2.2. Reinforcement learning-based Beam hopping

BH has been extensively studied as a key resource-allocation technique to cope with spatially and temporally heterogeneous ground traffic. In LEO satellite systems, multi-satellite cooperation becomes a central challenge due to overlapping coverage and rapidly varying link conditions.

Lin *et al.* [1] proposed a framework that decomposes the multi-satellite BH problem into two core components: load balancing and interference avoidance. Exploiting the multiple-coverage property of LEO constellations, the framework first distributes hotspot traffic across several satellites load balancing. Each satellite then independently designs a BH pat-

tern for its assigned cells to satisfy per-cell demand while maintaining spatial isolation among simultaneously illuminated beams. Finally, the sequence of per-satellite BH patterns is jointly scheduled to mitigate inter-satellite interference avoidance. While this decomposition provides a clear problem structure, the approach in [1] relies on predefined heuristics to solve the subproblems sequentially, limiting adaptability to non-stationary traffic and unpredictable network dynamics.

Complementary to the above line, Xie *et al.* propose a DRL approach for multi-LEO BH with joint beam scheduling and power allocation [2]. Their method formulates a hybrid action space—discrete beam-illumination patterns and continuous per-beam power—and employs PPO with two policy heads sharing a common feature extractor. The objective jointly optimizes long-term throughput and long-term cumulative average delay under full-frequency reuse, explicitly accounting for both intra- and inter-satellite interference in the SINR model. Simulation results show notable gains over heuristic baselines (e.g., up to 8.9% higher throughput and up to 69.2% lower LTCAD), highlighting the potential of DRL for traffic-driven BH in time-varying environments. Nonetheless, the learning agent is effectively centralized and assumes access to global network state (e.g., via NOCC), and the model does not explicitly consider GW selection or feeder-link switchover effects on service continuity.

To address these limitations, this paper adopts a multi-agent reinforcement learning MARL formulation. We model each satellite as an autonomous agent that observes local network states (e.g., per-cell de-

mand, channel/interference conditions, and GW status) and coordinates with other agents to learn a cooperative BH policy. The learned policy simultaneously optimizes interference avoidance and gateway load balancing, thereby improving service SINR and alleviating traffic bottlenecks. In contrast to the heuristic decomposition in [1] and the centralized DRL paradigm in [2], our MARL framework natively supports distributed decision-making under partial observability and explicitly couples BH with GW selection and feeder-link switchover, which are critical for end-to-end performance in practical NTN deployments.

Chapter 3. Proposed multi-agent reinforcement learning-based link scheduling algorithm

3.1. System model

3.2. Problem formulation

3.3. Proposed algorithm

3.3.1 MDP modeling

3.3.2 Proposed dueling DQN algorithm

Chapter 4. Performance evaluation

4.1. Simulation setup

4.2. Key Performance Indicators

4.3. Results and Discussion

Chapter 5. Conclusion

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